2022 DAYS on DIFFRACTION (DD), pp. 88–94 979-8-3503-4546-9/22/$31.00 © 2022 IEEE

Diabolical points and Rayleigh-wave propagation

Peter G. Malischewsky
Institute of Geosciences, Friedrich-Schiller University Jena, Germany;
e-mail: p.mali@uni-jena.de

Eigenvalue problems with "peculiar properties" were studied in a famous paper by von Neumann and Wigner. The phenomenon of diabolical points is placed into an extended context and is connected for the first time with Rayleigh-wave propagation. The associated appearance of osculation, avoided crossing or repulsion is discussed by using simple models.

Everything is connected with everything.
Alexander von Humboldt (1769–1859)

1 Introduction

We present here an interesting phenomenon, which is well-known in quantum physics, crystal optics, physical chemistry, and acoustics but less known in theory of Rayleigh waves, namely the occurrence of *diabolical points*. This phenomenon is closely connected with "avoided crossings" and these ones are called a fascinating phenomenon in the book by Lax [1]. It is denoted in different disciplines in different ways: namely "avoided crossing" in mathematics, quantum theory, and acoustics, "curve veering" in vibration theory, "osculations" in seismology, and "repulsion" in NDT (Non-Destructive Testing). Among recent works on osculations, an interesting essay by Forbriger [2] cogitates at length about the topic of avoided crossings and carries out a thorough review of previous works on osculating curves in seismology. On the one hand, this phenomenon is omnipresent in many waveguide problems, but on the other hand it requires a constellation of parameters satisfying miscellaneous constraints. Fig. 1 is an example of dispersion curves with many avoided crossings for Rayleigh and Love waves in a complicated mantle model after Kennett [3]. It is something like an "entrance gate" for the following considerations and illustrates what we are talking about.

Eigenvalue problems with "peculiar properties" were studied in the famous paper by von Neumann and Wigner [4]. They point out that it is important for many problems of quantum mechanics to investigate the change of eigenvalues and eigenfunctions in continuously modifying one or more parameters. Just this is crucial for the topic under consideration. It is well-known that in quantum mechanics many dynamical processes are described by avoided level crossings. In accordance with Keck et al. [5], they typically appear in the form of a matrix Hamiltonian, where the matrix elements depend on parameters. These avoided level crossings occur when a single parameter is varied. However, true crossings require the variation of more parameters, two in the case of 2×2 matrices, which constitutes the celebrated "diabolical crossing" of the energy levels. Teller [6] showed that the surfaces representing energy levels E in the space of E and the two parameters X, Y are connected at degeneracies like the two sheets of a double cone: a *diabolo*. To emphasize this conical geometry in the neighbourhood of degeneracies, Berry and Wilkinson [7] refer to the connections as "diabolical points". Diabolo comes from a Chinese toy with this name and shape and a pellet from an air gun is also sometimes referred to as diabolo. Depending on the context, this point is also denoted as *Dirac point*. According to Seyranian et al. [8], two important cases are distinguished: the diabolical points (DPs) and the exceptional points (EPs). From the mathematical point of view, DP is a point where the eigenvalues coalesce (become double), while corresponding eigenvectors remain different (linearly independent); and EP is a point where both eigenvalues and eigenvectors merge. A prominent example of a DP (see Fig. 2) can be found in the famous textbook of Landau and Lifschitz [9]. That concerns the so-called electron terms (U) of polyatomic molecules in dependence on 2 internuclear distances r_1 and r_2. Because of geometrical reasons, a *double conus* occurs with the *diabolical point* in the middle. The occurrence of this point is necessarily connected with degeneration of the eigenvalues in this point as mentioned above.

The crossing of eigenvalue surfaces (energy levels) can exhibit a number of characteristic forms beside the double conus. Fig. 3 shows an example after Kirillov [10].

We mention that in physics the double cone may well be traced back to the works of Hamilton [11] who expounded on an interesting physical effect as-

2022 Days on Diffraction (DD) | 979-8-3503-4546-9/22/$31.00 ©2022 IEEE | DOI: 10.1109/DD55230.2022.9961028

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

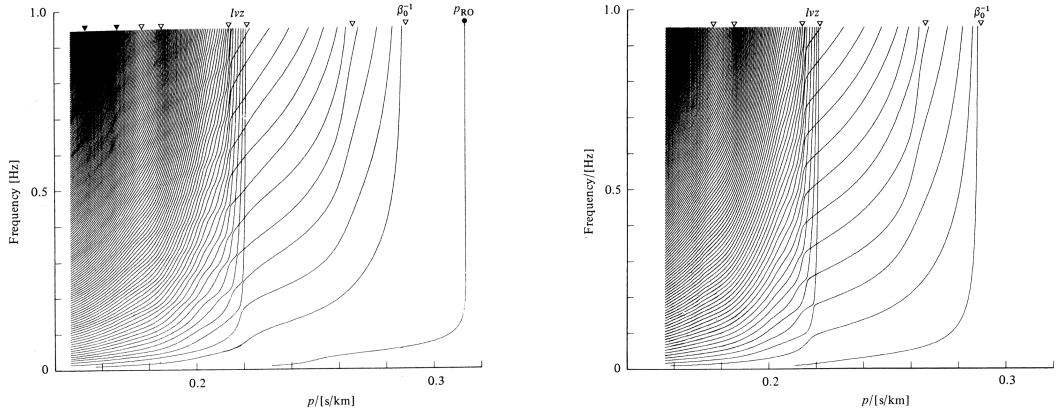


Figure 1: Dispersion curves frequency over slowness for the mantle model T7 after Kennett [3]; left: Rayleigh wave dispersion, right: Love wave dispersion.

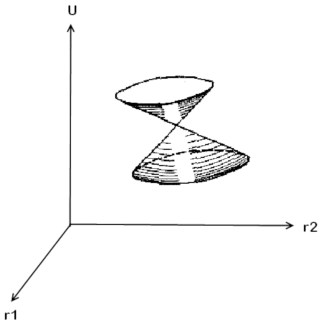


Figure 2: Example of a DP in Quantum Mechanics after Landau and Lifshitz [9].

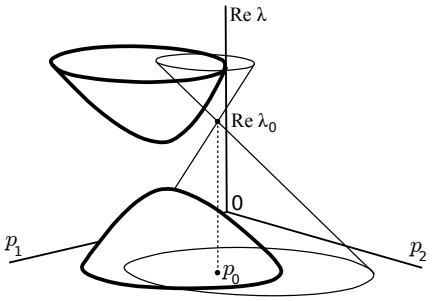


Figure 3: Unfolding of a DP due to complex perturbation after Kirillov [10].

sociated with coincident eigenvalues referred to as *conical refraction*.

2 Eigenvalue equations for Rayleigh and Love waves, possible degenerations and diabolical points

Let us start with two simple models: Layer with Fixed Bottom (LFB) and Layer Over Halfspace (LOH), see Fig. 4.

LFB is a closed waveguide and LOH an open one. A Cartesian coordinate system is introduced: while the horizontal x_1-axis is situated on the upper surface and points into the direction of wave propagation, the vertical x_3-axis goes through the cross section of the waveguide and x_2 is the transversal direction. The thickness of the waveguide and the layer, respectively, is H. The upper surface of these waveguides is assumed to be stress-free, i.e. with the corresponding stress tensor components

$$S_{13}(0) = S_{23}(0) = S_{33}(0) = 0, \tag{1}$$

and the motion vanishes on the bottom for LFB, i.e. the displacements $U_i(x_3)$ $(i = 1, 2, 3)$ vanish:

$$U_1(H) = U_2(H) = U_3(H) = 0. \tag{2}$$

These are the boundary conditions. For model LOH, the usual continuity conditions hold. We assume an ideal elastic medium and harmonic motion. The Navier equation leads to a coupled system of differential equations for Rayleigh waves (see, e.g., Malischewsky [12]):

$$\begin{aligned} \beta^2 U_1'' + ik\big(\alpha^2 - \beta^2\big)U_3' + \big(\omega^2 - k^2\alpha^2\big)U_1 = 0, \\ \alpha^2 U_3'' + ik\big(\alpha^2 - \beta^2\big)U_1' + \big(\omega^2 - k^2\beta^2\big)U_3 = 0, \end{aligned} \tag{3}$$

and a separated equation for Love waves:

$$U_2'' - \big(k^2 - k_\beta^2\big)U_2 = 0, \tag{4}$$

which adopts the form of a singular Sturm–Liouville problem for a layered half-space $x_3 \in [0, \infty)$:

$$LU_2 = \Lambda U_2, \tag{5}$$

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

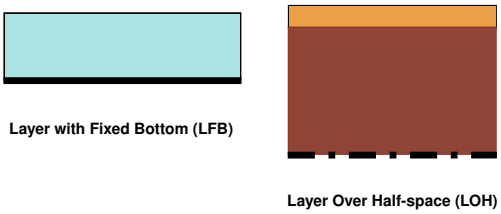


Figure 4: Models under consideration: left: LFB; right: LOH.

where $\Lambda = k^2$ and

$$L = \frac{1}{\mu(x_3)}\left[\frac{d}{dx_3}\left(\mu(x_3)\frac{d}{dx_3}\right) + \rho(x_3)\omega^2\right]. \quad (6)$$

The variables in Eq. (3)–(5) are defined as follows: α and β are P- and S-velocity, respectively, μ and λ are Lamé parameters, ρ is density, ω is the circular frequency, k is wave number, and $k_\beta = \omega/\beta$. For LFB, $x_3 \in [0, H]$ but for LOH $x_3 \in [0, \infty)$. The boundary condition (2) has to be modified for LOH by setting $H \to \infty$. Because of (5), the eigenvalues of Love waves will never degenerate. It has to be noted that the eigenvalue problem (1)–(3) is not a standard one, because the eigenvalue k enters linearly and quadratically. However, following Kirrmann [13] and Besserer and Malischewsky [14], this problem can be reformulated by introducing a column motion vector $\vec{w}$,

$$\vec{w} = \begin{pmatrix} U_1 \\ U_3 \\ S_{11} \\ S_{13} \end{pmatrix}, \quad (7)$$

together with the special matrix operator $\mathcal{L}$ and the eigenvalue equation

$$\mathcal{L}\vec{w} = i\,k\,\vec{w} \quad (8)$$

is convenient for a subsequent functional analytical consideration concerning completeness. For reasons of space, we do not specify here the complicated matrix $\mathcal{L}$ which can be found in the literature given.

The secular equation for Rayleigh waves in LFB can be written according to Tran Thanh Tuan [15] in the form:

$$\begin{aligned}\Delta(C,F) &= A_0(C) \\ &+ B_0(C)\sin\left(\frac{2\pi F g_\alpha}{C}\right)\sin\left(\frac{2\pi F g_\beta}{C}\right) \\ &+ C_0(C)\cos\left(\frac{2\pi F g_\alpha}{C}\right)\cos\left(\frac{2\pi F g_\beta}{C}\right) = 0 \quad (9)\end{aligned}$$

with $g_\alpha = \sqrt{\gamma C^2 - 1}$, $g_\beta = \sqrt{C^2 - 1}$, $\gamma = \beta^2/\alpha^2$.

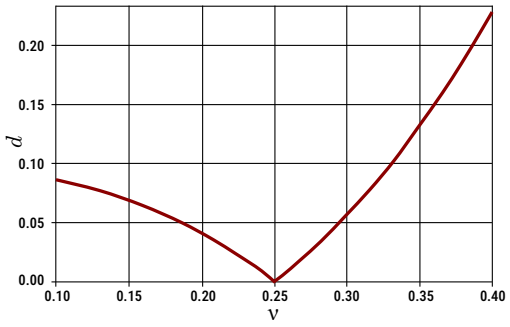


Figure 5: Distance d between mode 1 and 2 depending on ν.

The dimensionless frequency F and phase velocity C, respectively, are defined by

$$F = H/\lambda_\beta, \quad C = c/\beta, \quad (10)$$

where λ_β is shear wavelength and c is phase velocity. The auxiliary functions $A_0(C)$, $B_0(C)$, $C_0(C)$ are given by

$$\begin{aligned} A_0(C) &= -4g_\alpha g_\beta\left(C^2 - 2\right), \\ B_0(C) &= C^4(4\gamma + 1) - 4C^2(\gamma + 2) + 8, \\ C_0(C) &= -g_\alpha g_\beta\left(C^4 - 4C^2 + 8\right). \end{aligned} \quad (11)$$

By using (4) the corresponding secular equation for Love waves in LFB is

$$\cos\left(2\pi F\sqrt{1 - 1/C^2}\right) = 0. \quad (12)$$

The investigation of possible degenerations of eigenvalues is by far not trivial. The celebrated von Neumann – Wigner theorem describes the probability that a finite-dimensional matrix has a degenerate eigenvalue. This probability is lower than might be expected: the codimension of the set of matrices with a double eigenvalue is always greater than one. It follows for a time-dependent Hamiltonian, the probability that two energy curves intersect is extremely low, and this phenomenon is called level repulsion (Demkov and Kurasov [16]). Kausel et al. [17] carefully studied this situation in the context of elastic waveguides. We discuss it more thoroughly insofar it concerns the models LFB and LOH. The crucial parameter is Poisson's ratio ν, which can be written with the ratio γ as

$$\nu = \frac{1 - 2\gamma}{2(1 - \gamma)}. \quad (13)$$

By analyzing an impedance matrix it was found out by Kausel et al. [17] that the eigenvalues for

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

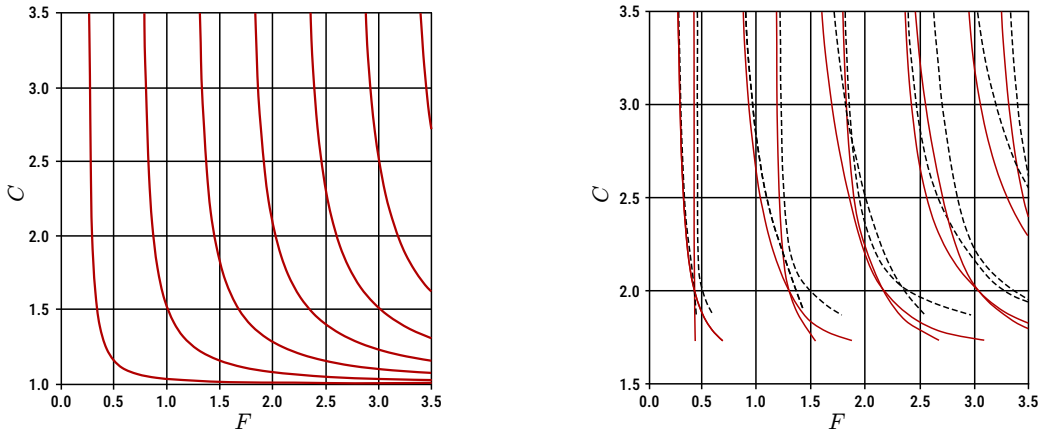


Figure 6: left: Dispersion curves of Love waves for LFB; right: Dispersion curves of Rayleigh waves for LFB, red: $\nu = 0.25$, dashed: $\nu = 0.3$.

LFB degenerate, i.e. the dispersion curves or spectral lines cross, only then, when ν is either exactly 1/4 (case 1) or is another special rational fraction (cases 2 and 3) within the interval $0 < \nu < 1/3$. There are infinitely many double roots within this interval. No double roots exist for LFB when Poisson's ratio is either an irrational number, or a rational number that does not satisfy either of the two equations according to case 2 and 3. We specify here only case 1, omit the formulas for cases 2 and 3, but give some examples for ν in Table 1,

$$\begin{aligned} &\text{case 1}: \quad \nu = 1/4, \quad C = 2, \\ &\qquad F = \sqrt{3}/2(n + 1/2), \quad n = 0, 1, 2, \ldots \end{aligned} \tag{14}$$

Fig. 5 shows the dependence of distance d (in non-dimensional units) between mode 1 and 2 from ν for LFB. For $\nu = 0.25$, d is zero, i.e. a case of crossing or degeneration.

This close connection between degenerations of eigenvalues and Poisson's ratio is noteworthy and interesting. By using a dispersion equation for LFB formulated via ray theory, Tran Thanh Tuan [15] succeeded in obtaining similar results, however the solution 3 was overlooked there.

Table 1: Selected Poisson's ratios for case 2 and 3.

case 2		case 3	
ν (as fraction)	ν	ν (as fraction)	ν
1/18	0.055556	1/99	0.010101
1/14	0.071429	1/9	0.111111
1/10	0.100000	13/63	0.206349
1/6	0.166667	37/135	0.274074
3/14	0.214286	13/45	0.288889
3/10	0.300000	61/189	0.322751

For LFB and Rayleigh waves, we observe strong osculations of spectral lines for nearby Poisson ratios according cases 1–3. Also LOH shows an abundance of osculations, however, Kausel et al. [17] formulate the strong conjecture that LOH and beyond that all layered half-spaces never exhibit crossings of spectral lines. This is strictly valid for Love waves as well, but LFB and LOH have no osculations for them. Their occurrance requires more complicated models (Levshin [18]). The following figures illustrate this behaviour by means of a few selected examples using (9) and (12) and the secular equation for LOH not specified here.

Fig. 6 concerns LFB and shows the dispersion curves for Love and Rayleigh waves with $\nu = 0.25$ and 0.30: no osculations for Love waves, but a lot of them together with crossings for Rayleigh waves. Note that beside the crossings for $\nu = 0.25$ a crossing between the 5th and 6th mode (after case 2) for $\nu = 0.3$ occurs at $F \approx 1.87$, $C \approx 2.828$.

For contrast, Fig. 7 shows dispersion curves of Rayleigh waves for LOH with $\nu = 0.25$ in layer and half-space, shear contrast $r_s = 0.25$ and density contrast $r_d = 0.75$ between layer and half-space. There are no crossings but several osculations.

As already noted, the formation of a diabolical point requires the crossing (degeneration) of spectral lines and variation of 2 parameters, namely in our case frequency F and Poisson's ratio ν. Fig. 8 shows for the first time a diabolical point

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

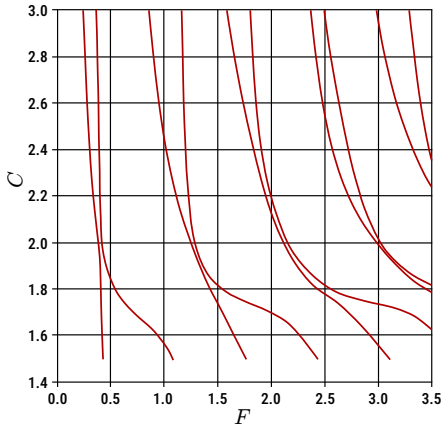


Figure 7: Dispersion curves of Rayleigh waves for a special LOH model.

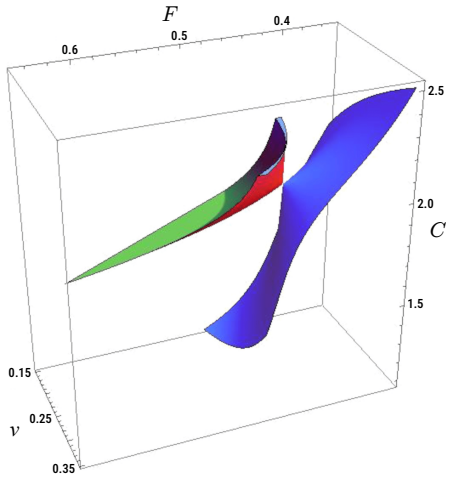


Figure 8: 3D-picture of the function $C = C(F, \nu)$ for LFB: fundamental mode (blue, right) and higher mode (red/green, left).

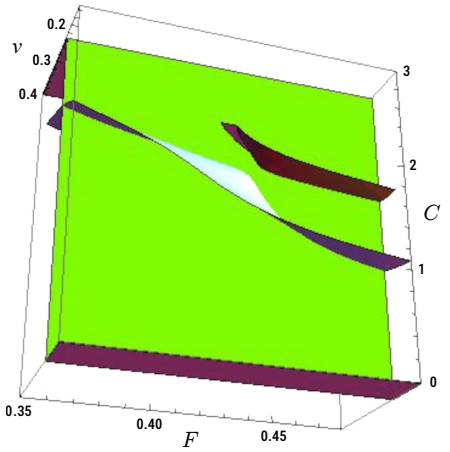


Figure 9: Visualization of an avoided crossing by slicing Fig. 8 with the plane $\nu = 0.23$.

for Rayleigh waves in LFB by presenting the phase velocity of the fundamental and first higher mode depending on F and ν. The double conus is clearly seen with the vertex given by case 1 and formula (14) with $n = 0$. The surfaces for the fundamental and first higher mode, respectively, touch for $C = 2$, $F = \sqrt{3}/4$, $\nu = 0.25$ according to (14).

It should be noted that there is an abundance of diabolical points in conformity with Table 1. However, most of them have no practical interest, because they occur at high frequencies and mode numbers. Berry and Wilkinson [7] say that the avoided crossings are simply approximately double-hyperbola curves obtained by slicing the diabolo near its vertex. This is demonstrated in Fig. 9, where the diabolo of Fig. 8 is sliced by a plane $\nu = 0.23$ with the effect that an avoided crossing becomes visible.

A sideline investigation of the author for the critical point $C = 2$, $F = \sqrt{3}/4$ suggested that this point is really a DP (compare with Seyranian et al. [8]) and not an EP. But this statement requires further substantiation in future.

3 Outlook and concluding remarks

The existence of diabolical points in the theory of Rayleigh waves is a prime example for the strong unifying element of mathematics between different disciplines of science. While the benefit remains unclear up to now, it holds some fascination without doubt and will trigger deeper theoretical investigations in this area. However, the investigation of avoided crossing is already today very important and interesting for seismology and NDT. Because of the abundance of material, a few references only can be given here. In seismology, the osculations can lead to a misidentification of recorded Rayleigh modes with the consequence of an incorrect inversion for a subsurface model (see, e.g., [19, 20]). Following Lugovtsova et al. [21], Lamb wave mode repulsion may be applied for the development of NDT-methods to characterize the mechanical strength of adhesive bonds. Glushkov and Glushkova [22] demonstrate how the phenomenon of ZGV points of Lamb waves with the associated mode repulsion can be used for NDT-material characterization.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

We have seen that Poisson's ratio is crucial for certain aspects of wave propagation. That is why a recent paper about the elegant determination of ν using ZGV Lamb waves is mentioned (Watzl et al. [23]).

Finally we state that the diabolical point is really diabolic (nomen est omen), because the mathematics we are familiar with breaks down in it. But certain analytical laws may be found in its surrounding by series expansion.

Acknowledgments

The author expresses his sincere gratitude to all colleagues and friends from PDMI and the University of St. Petersburg for the cooperation and friendship over more than half a century. In this connection I would like also to remember the deceased colleagues and friends L. A. Molotkov and T. B. Yanovskaya. Furthermore, I express my sincere thanks to all Russian colleagues and friends in Moscow and elsewhere. My father Sigismund Malischewsky (1896–1984) had always a close and respectful relationship with St. Petersburg, where he served in the Tsar's army. I devote this article to the memory of my aunt Sophie Biberger née Malischewsky (1900–1974) and uncle Albert Biberger (1889–1972), who were like my second parents and from whom I have learnt a lot.

References

[1] Lax, P. D., 2007, *Linear Algebra and its Applications*, Wiley, Hoboken/New Jersey.

[2] Forbriger, T., 2006, Some thoughts on osculation and dispersion curves, Internet publication at `https://publikationen.bibliothek.kit.edu/1000037033/2855784` (in German).

[3] Kennett, B. L. N., 2009, *Seismic Wave Propagation in Stratified Media*, ANU Press, Canberra.

[4] von Neumann, J., Wigner, E., 1929, On peculiar discrete eigenvalues, *Physikalische Zeitschrift*, Vol. **30**, pp. 465–467 (in German).

[5] Keck, F., Korsch, H. J., Mossmann, S., 2003, Unfolding a diabolical point: a generalized crossing scenario, *Journal of Physics A: Mathematical and General*, Vol. **36**, pp. 2125–2137.

[6] Teller, E., 1937, The crossing of potential surfaces, *Journal of Physical Chemistry*, Vol. **41**, pp. 109–116.

[7] Berry, M. V., Wilkinson, M., 1984, Diabolical points in the spectra of triangles, *Proceedings Royal Society of London A*, Vol. **392**, pp. 15–43.

[8] Seyranian, A. P., Kirillov, O. N., Mailybaev, A. A., 2005, Coupling of eigenvalues of complex matrices at diabolic and exceptional points, *Journal of Physics A: Mathematical and General*, Vol. **38**, pp. 1723–1740.

[9] Landau, L. D., Lifshitz, E. M., 1981, *Course of Theoretical Physics. Vol. 3: Quantum Mechanics*, Elsevier, Amsterdam.

[10] Kirillov, O. N., Mailybaev, A. A., Seyranian, A. P., 2005, Unfolding of eigenvalue surfaces near a diabolic point due to complex perturbation, *Journal of Physics A: Mathematical and General*, Vol. **38**, pp. 5531–5546.

[11] Hamilton, W. R., 1833, On a general method of expressing the paths of light, and of the planets, by the coefficients of a characteristic function, *Dublin University Review and Quarterly Magazine*, Vol. **1**, pp. 795–826.

[12] Malischewsky, P., 2020, Surface waves, In: *Encyclopedia of Continuum Mechanics*, pp. 2393–2402, Springer, Berlin.

[13] Kirrmann, P., 1995, On the completeness of Lamb modes, *Journal of Elasticity*, Vol. **37**, pp. 39–69.

[14] Besserer, H., Malischewsky, P. G., 2004, Mode series expansions at vertical boundaries in elastic waveguides, *Wave Motion*, Vol. **39**, pp. 41–59.

[15] Tran Thanh Tuan, 2009, *The ellipticity (H/V-ratio) of Rayleigh surface waves*, PhD dissertation, Friedrich-Schiller University Jena, Institute of Geosciences, Germany, `https://d-nb.info/993363504/34`.

[16] Demkov, Yu. N., Kurasov, P. B., 2007, Von Neumann – Wigner theorem: level repulsion and degenerate eigenvalues, *Theoretical and Mathematical Physics*, Vol. **153**, pp. 1407–1422.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.

[17] Kausel, E., Malischewsky, P., Barbosa, J., 2015, Osculations of spectral lines in a layered medium, *Wave Motion*, Vol. **56**, pp. 22–42.

[18] Levshin, A. L., 1973, *Surface and Channel Seismic Waves*, Nauka, Moscow (in Russian).

[19] Forbriger, T., 2003, Inversion of shallow-seismic wavefields: I. Wavefield transformation, *Geophysical Journal International*, Vol. **153**, pp. 719–734.

[20] Forbriger, T., 2003, Inversion of shallow-seismic wavefields: II. Inferring subsurface properties from wavefield transforms, *Geophysical Journal International*, Vol. **153**, pp. 735–752.

[21] Lugovtsova, Y., Johannesmann, S., Henning, B., Prager, J., 2019, Analysis of Lamb wave mode repulsion and its implications to the characterisation of adhesive bonding strength, *Proceedings of Meetings on Acoustics (POMA)*, Vol. **38**, pp. 1–5.

[22] Glushkov, E. V., Glushkova, N. V., 2021, Multiple zero-group velocity resonances in elastic layered structures, *Journal of Sound and Vibration*, Vol. **500**, 116023.

[23] Watzl, G., Kerschbaummayr, Ch., Schagerl, M., Mitter, Th., Sonderegger, B., Grünsteidl, C., 2022, In situ laser-ultrasonic monitoring of Poisson's ratio and bulk sound velocities of steel plates during thermal processes, *Acta Materialia*, Vol. **235**, 118097.

Authorized licensed use limited to: University of Shanghai For Science and Technology. Downloaded on September 10,2026 at 07:54:18 UTC from IEEE Xplore. Restrictions apply.